Cryopreservation of gonadal tissue

Ovarian tissue cryopreservation (OTC) involves harvesting and freezing ovarian tissue, thus preserving oocytes in primordial follicles located in the ovarian cortex. The ovarian tissue can subsequently be auto transplanted after gonadotoxic treatment has ended.

A low-quality, limited body of evidence suggests that OTC and transplantation have the potential to restore ovarian function and may result in preserved fertility in patients who have undergone gonadotoxic cancer treatment. Limitations to the body of evidence include an evidence base composed of 2 poor-quality cohort studies, 6 poor-quality single-arm studies, and 1 very-poor-quality cross-sectional study. Better quality prospective studies ensuring that all patients are followed after receiving transplantation would provide better assurance that the effects of OTC and subsequent transplantation on fertility and pregnancy outcomes reported herein are representative.

Cryopreservation, storage and thawing of testicular tissue is considered unproven in the treatment of infertility (ASRM, 2014).

Cryopreservation of testicular tissue in higher risk prepubertal males is feasible, but still experimental. Female fertility preservation is more complex, requires more invasive procedures, and can delay initiation of treatment due to the requirement for hormone stimulation of follicles prior to harvesting (Romao & Lorenzo, 2017).

Co-culture of embryos

Studies describe different techniques of co-culture, but no standardized method of co-culturing has been defined. Further studies are necessary to support the effects of co-culture on clinical outcomes.

Co-culture is a process by which embryos develop on culture containing cells grown from the patient's own endometrium (uterine lining). The endometrial cells collected by the physician are grown in the Embryology Laboratory. These cells are thawed prior to oocyte retrieval. Then, the embryos are placed with the cultured endometrial cells to support embryonic development (Weill Cornell Medicine, 2023).

Le Saint et. al. (2019) performed an interventional, randomized, double-blind study that took place at Clinique Ovo from March 2013 to October 2015 and included 207 healthy patients undergoing an IVF of which 71 were excluded before randomization. On the previous cycle, all participants underwent an endometrial biopsy at D5 to D7 post-ovulation, following which the endometrial cells were prepared for (AECC) autologous endometrial cell co-culture.

The data demonstrated that AECC significantly increased the incidence of good-quality blastocysts compared with culture in conventional media (42.6% versus 28.4%, $P < 0.001$). No significant differences were found in pregnancy and live birth rates.